ADVAITH R PAI

Kochi, Kerala

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EXPERIENCE

DRDO Naval Physical and Oceanographic Laboratory | Kochi
Research Intern **June 2026 - Present**
Research Intern **September 2025 - December 2025**

Architected a scalable deep learning framework for automated passive sonar classification, directly enabling NPOL's underwater acoustics surveillance capabilities.

- Built the initial end-to-end audio pipeline processing **10,000+ sonar samples** into Mel-spectrograms, reducing preprocessing latency by **35%**
- Developed the foundational hybrid **DWSTr (Depthwise Separable CNN + Transformer)** model, achieving a benchmark **97.17% accuracy** across **12 vessel categories** by implementing square root class-weight smoothing to resolve severe dataset imbalances
- **Engineered a memory-efficient data pipeline** to systematically chunk high-resolution audio files into compressed `.npz` tensors, mitigating cloud Out-of-Memory (OOM) bottlenecks during scale-up
- **Scaled the DWSTr deep learning framework** to process the massive 47-hour DeepShip real-world acoustic dataset, targeting the classification of 265 distinct vessels across 4 commercial categories
- Presented the model to senior leadership, facilitating high engagement Q&A to gather actionable feedback for iterative model enhancements.

Reference: Pratheek Sureshkumar · Scientist · NPOL, DRDO Kochi

PROJECTS

SERVD - AI-Powered Kitchen Assistant **February 2026**

- Developed full-stack AI recipe platform integrating **Gemini Vision AI, TheMealDB, and Razorpay** for automated ingredient recognition and recipe generation via Next.js and Strapi
- Engineered the backend with **PostgreSQL**, implementing pantry ingestion workflows (image scanning + manual entry), secure Clerk authentication, and tier-based premium access controls
- **Tech Stack: Next.js, Tailwind CSS, Strapi, PostgreSQL/SQLite, Google Gemini AI, Clerk Auth, Razorpay, Arcjet**

SELECTRA - AI Resume Analyzer **September 2025**

- Built AI-powered resume screening platform using **Claude Opus 4.1** and **PDF.js**, processing resumes to generate **ATS compatibility scores** and actionable improvement recommendations across 5 evaluation criteria
- Integrated **Puter.js cloud infrastructure** (file storage, KV database, AI API) with **Zustand** state management, enabling persistent resume tracking and **60% improved user retention**
- **Tech Stack: React Router v7, TypeScript, Tailwind CSS, Puter.js, PDF.js, Zustand**

TECHNICAL SKILLS

Languages:	Python, TypeScript, SQL
Software Development:	REST APIs, Full-Stack Development, Git, GitHub
Frameworks & AI Tools:	React, Next.js, Tailwind CSS, TensorFlow, Librosa, Scikit-learn
Databases & Cloud:	PostgreSQL, SQLite

EDUCATION & ACHIEVEMENTS

Muthoot Institute of Technology & Science
Bachelor of Technology in Computer Science - AI Engineering CGPA: 7.56/10

GitHub Leader: Gained **1000+ followers** through consistent contributions & high-quality projects

Consistent Programming: Solved **300+ problems** on LeetCode, demonstrating strong Data Structures and Algorithms proficiency